

Project Executable Files

Date	6-June 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

/HDI-Predictor

/data - HDI dataset (CSV) used for training

/notebooks - Jupyter notebooks for EDA and model experimentation

/model - Trained model file (hdi_model.pkl) and scaler

/static - CSS, JS, and image assets

/templates - HTML templates (index.html, result.html)

app.py - Flask application with prediction route

train_model.py - Script to preprocess data and train the ML model

requirements.txt - Python dependencies (Flask, scikit-learn, pandas, numpy, matplotlib, seaborn)

README.md - Setup and run instructions

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	To be updated after deployment (e.g., Render/Heroku link)
Login Credentials (Demo)	Not applicable — application is open access, no login required
Platform / Hosting Provider	Render / Anaconda-based local hosting
Repository Link	To be updated with the team's GitHub repository link
Demo Video Link	To be updated after recording the final demo walkthrough

Step 4: Run Instructions

Pre-requisites: Python 3.8+, Anaconda (recommended), pip, and the libraries listed in requirements.txt (scikit-learn, numpy, pandas, matplotlib, seaborn, Flask).

Step 1: Clone the repository: `git clone https://github.com/Parneeth/HDI-project`

Step 2: Create and activate a virtual/conda environment

Step 3: Install dependencies: `pip install -r requirements.txt`

Step 4: (If not already trained) Run: `python train_model.py` to generate the model file

Step 5: Start the application: `python app.py`

Step 6: Open browser at hdi-project-xb3w.onrender.com/ and enter country indicators to predict the HDI category

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Model accuracy depends on the completeness of the training dataset	Using a cleaned, well-known HDI dataset; documented as a future improvement area
2	No authentication on the web app	Acceptable for demo scope; planned for future scalability phase
3	Local deployment only during development	Public deployment link to be added before final submission